

AYUSH SAHOO

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Summary

Full-stack developer with hands-on experience building and deploying **production-grade web applications** using **React**, **Node.js**, **Express.js**, and **MongoDB**. Shipped **three end-to-end applications** from architecture through **public production deployment** spanning real-time communication, algorithmic trading systems, and distributed version control.

Education

Sir M. Visvesvaraya Institute of Technology

Bachelor of Engineering in Information Science

Sep 2023 – Jul 2027

CGPA: 8.38/10.0

Technical Skills

Languages: Java, JavaScript, C, SQL

Frontend: React.js, HTML5, CSS3, Chart.js, Recharts

Backend: Node.js, Express.js, REST APIs, Socket.io, WebSockets

Databases: MongoDB, MySQL

Version Control & Deployment: Git, GitHub, Vercel, Render

Technologies: WebRTC, JWT, bcrypt, Cloudinary, TURN/STUN, ICE

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Projects

VidMeet – Real-Time Video Conferencing Platform | *React, WebRTC, Socket.io, Node.js* [GitHub](#) | [Live](#)

- Engineered a real-time video conferencing platform using **WebRTC** and **Socket.io**, enabling low-latency peer-to-peer audio/video communication, synchronized chat, screen sharing, and dynamic meeting generation.
- Achieved **99.9% connection reliability** across restrictive corporate and educational networks by implementing **TURN/STUN infrastructure** and configuring ICE candidates over ports **80** and **443**.
- Eliminated video stream flickering during real-time interactions by isolating media elements into reusable **React** components and optimizing rendering with **useRef**-based state management.
- Secured meeting access through **JWT authentication**, host-controlled waiting rooms, active participant tracking, meeting history management, and real-time network quality monitoring using WebRTC connection statistics.

GitVerse – Distributed Version Control System | *React, Node.js, Express.js, MongoDB Atlas* [GitHub](#) | [Live](#)

- Engineered a custom **Node.js CLI** supporting **8+ version control operations** (including **init**, **remote add**, **push/pull**, and **revert**), enabling bidirectional cloud synchronization and **.gitverseignore**-based file exclusion.
- Designed and implemented a **JWT-authenticated Express.js backend** using **MongoDB Atlas** and **Cloudinary**, enabling secure cloud synchronization, storage, and retrieval of repository files with **public/private access controls**.
- Developed a **React-based dashboard** featuring **public repository discovery**, **commit history visualization**, **file-level commit inspection**, and a dynamic **contribution heatmap** for tracking project evolution and developer activity.
- Resolved **Linux-specific case-sensitivity build failures** blocking production deploys and implemented **environment-aware API routing** across **Vercel** (frontend) and **Render** (backend), eliminating recurring deployment failures across **dev/staging/prod**.

StockFlow – Real-Time Trading & Order Execution Engine | *Node.js, Socket.io, Gemini AI* [GitHub](#) | [Live](#)

- Engineered a full-stack **virtual stock trading platform** supporting portfolio management and simulated trading with **INR 10,00,000 virtual capital**, by integrating live market data and portfolio analytics using **Node.js**, **React**, and **MongoDB**.
- Reduced order evaluation complexity from **O(N)** to **O(logN)** by implementing an **algorithmic limit-order execution engine** using **Min-Heap** and **Max-Heap** priority queues for automated trade execution.
- Enabled real-time market monitoring and order execution by building a bi-directional **WebSocket** data pipeline with **Socket.io** and **Yahoo Finance**, streaming live stock prices at 10-second intervals.
- Integrated **Gemini AI** to generate real-time buy/sell insights from live market data and implemented a **persistent transaction ledger** with a **full audit trail** for every simulated trade, secured with **JWT authentication**.

Certifications & Achievements

- GenAI with JavaScript – ChaiCode [View Certificate](#)
- Full Stack Web Development (Delta) – Apna College [View Certificate](#)
- Data Structures & Algorithms in Java (Alpha) – Apna College [View Certificate](#)
- Solved 350+ Data Structures & Algorithms problems on LeetCode.